Diskrete Mathematik – Woche 8 (Suggested)

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Divisors, Division with Remainders and gcd

Problem 1 ⋆

w08 1.3, 1.6, 1.7

Prove or disprove for all $a, b, c \in \mathbb{Z}$.

1. $a \mid b \wedge b \mid a \Rightarrow a = b$
2. $a \mid bc \Rightarrow a \mid b \vee a \mid c$
3. $a \mid b \Rightarrow |a| \leq |b|$

Problem 2 ⋆

w08 2.1 (negative cases), 2.4

1. Determine q and r from Theorem 4.1 for $(a, d) \in \{(-17, 5), (17, -5), (-17, -5)\}$.
2. Prove or disprove for all $a \in \mathbb{Z}$ and $d \geq 1$: $R_d(-a) = d - R_d(a)$.

Problem 3 ⋆ ⋆

w08 4.1, 4.3

Let $a, b, k \in \mathbb{Z}$ with a, b not both 0, and $n \in \mathbb{N}$.

1. Prove: $\gcd(a + kb, b) = \gcd(a, b)$.
2. Determine $\gcd(n^2 + 1, n + 1)$.

Primes and Factorization

Problem 4 ⋆

w08 7.5

Find $u, v \in \mathbb{Z}$ with $56u + 15v = 1$.

Problem 5 ⋆ ⋆

w08 8.3

Prove or disprove for all $a, b, c \in \mathbb{Z}$: $a \mid c \wedge b \mid c \Rightarrow ab \mid c$. If the claim is false, repair it with an additional hypothesis.

Problem 6 ⋆ ⋆

w08 9.3

Prove: if $2^n - 1$ is prime ($n \geq 1$), then n is prime. Is the converse true?

Problem 7 ⋆ ⋆

w08 11.2

Prove: there are infinitely many primes p with $R_4(p) = 3$. Why does the same argument fail for $R_4(p) = 1$?

Congruences and Modular Arithmetic

Problem 8 ⋆

w08 12.1

Prove or disprove for all $a, b, c \in \mathbb{Z}$ and $m \geq 1$: $ac \underset{m}{\equiv} bc \wedge c \underset{m}{\not\equiv} 0 \Rightarrow a \underset{m}{\equiv} b$. If the claim is false, repair it with an additional hypothesis on c.

Problem 9 ⋆

w08 13.3

Compute $R_{100}(3^{2026})$.

Problem 10 ⋆ ⋆

w08 14.4

Prove that $x^3 + y^3 + z^3 = 2030$ has no integer solutions.

Problem 11 ⋆ ⋆

w08 16.4

Prove: $(p-1)! \underset{p}{\equiv} -1$ for every prime p.

Problem 12 ⋆ ⋆

w08 17.2

Find all solutions of $x \underset{6}{\equiv} 5$, $x \underset{10}{\equiv} 3$, $x \underset{15}{\equiv} 8$ with $0 \leq x < 30$, and justify that there are no others. For which $c \in \mathbb{Z}_{15}$ does the system with $x \underset{15}{\equiv} c$ instead of $x \underset{15}{\equiv} 8$ have a solution?

Problem 13 ⋆ ⋆

w08 18.1, 18.2

Alice and Bob run the Diffie-Hellman protocol (Figure 4.1) with $p = 23$ and $g = 5$.

1. Alice chooses $x_A = 6$ and Bob chooses $x_B = 15$. Compute y_A, y_B, k_{AB}, k_{BA}.
2. Eve sees only y_A and y_B. Compute k_{AB} the way Eve would for such a small p.